Rectangle Mania

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The problem

- Count axis-aligned rectangles whose four corners are all in the point set.
- Each pair forming a strictly up-right diagonal identifies one rectangle exactly once.

Intuition

- Two points forming an up-right diagonal define a rectangle if the other two corners exist.

Approach

- 1. Put points in a set
- 2. For each pair up and to the right
- 3. Check the other two corners
- 4. Count

Java solution

```java
public class RectangleMania {
    public static int rectangleMania(int[][] coords) {
        Set<String> set = new HashSet<>();
        for (int[] p : coords) set.add(p[0] + "," + p[1]);
        int count = 0;
        for (int[] a : coords) for (int[] b : coords)
            if (b[0] > a[0] && b[1] > a[1] && set.contains(a[0] + "," + b[1]) && set.contains(b[0] + "," + a[1])) count++;
        return count;
    }
}
```

Complexity

- Time: O(n^2)
- Space: O(n)

Recap

- Two points forming an up-right diagonal define a rectangle if the other two corners exist.
- Complexity: O(n^2)